

Physical Basis and Distinctiveness of Self-Supervised 4D-STEM Diffraction Clustering in Indomethacin Films: Robustness to Sample and Image Preprocessing

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Abstract

We ask what a self-supervised DINO model bases its clustering of 4D-STEM nanobeam electron diffraction (NBED) on, and whether the resulting partition could have been produced by a conventional 4D-STEM pipeline. Across three runs on two indomethacin (IMC) films (SI3 and SI4) and two image normalisations (linear vs. logarithmic intensity stretch), a CIF-free interpretability battery—representational probing, causal input ablation, and a classical-baseline comparison—gives one consistent answer: the clustering is driven by the overall scattered (diffracted) intensity together with the two-dimensional *low-q* ($q \lesssim 2.1 \text{ nm}^{-1}$, $d \gtrsim 4.7 \text{ Å}$) diffraction structure, and it is **not reproduced by any classical method**—virtual dark-field, azimuthal integration, PCA, NMF, or their combination all reach only $\text{ARI} \leq 0.18$ against the DINO map. The headline conclusion is robust to sample and preprocessing; the logarithmic stretch does not change *what* the model keys on but sharpens it, yielding more classes, far more linearly decodable physical factors, and a stronger coupling to the overall scattered intensity.

Keywords

4D-STEM, nanobeam electron diffraction, self-supervised learning, interpretability, unsupervised clustering, indomethacin

Introduction

Self-supervised clustering segments a 4D-STEM scan into structurally distinct regions without labels or a forward model, but the physical meaning—and the *novelty*—of the resulting classes is rarely established.^{1,2} Two questions matter before such a map is used: (i) which physical property separates the classes, and (ii) could a standard 4D-STEM analysis (a virtual detector, an azimuthal integration, a PCA/NMF decomposition) have produced the same segmentation? Here we answer both for a radial-gated, rotation-invariant DINO model^{1,3} on IMC NBED data, and test how robust the answer is across two films and two intensity normalisations.

Methods

Data and models. Three frozen runs were analysed: SI3 and SI4 (128×128 scans, 512×512 detector), with SI3 trained under both a linear (v_{max}) and a logarithmic intensity stretch. Active class counts are $K = 9$ (SI4), 12 (SI3

linear) and 14 (SI3 log). All analyses reuse the frozen checkpoints and saved inference (embeddings + hard labels).

Battery (CIF-free). (1) *Probing*: 5-fold CV ridge R^2 (decodability) and correlation ratio η^2 (separation) for the scattered intensity, crystallinity (1-D peak/halo), and spottiness (azimuthal variance). (2) *Causal ablation*: re-infer on perturbed inputs (scattered-intensity normalised, azimuthally averaged, low- q / high- q masked) and measure the adjusted Rand index (ARI) against the original map—low ARI means the model depends on that signal. (3) *Classical baselines*: cluster five classical feature sets (virtual dark-field, radial→PCA, pattern PCA, pattern NMF, and their combination) into the same K and score ARI/AMI against the DINO partition. No CIF or ACOM is required for any of this; orientation/phase analysis is deferred until indexing is available.

q calibration with the crop. The diffraction calibration is 0.0185 nm^{-1} per raw (512 px) pixel. Patterns are resized to 192 px (full field), then centre-cropped (120 px for SI4, 140 px for SI3) and resized again for the network. The low- q mask is applied in the uncropped 192-px frame ($0.0493 \text{ nm}^{-1}/\text{px}$), so its boundary is crop-independent at $q \simeq 2.1 \text{ nm}^{-1}$; the crop sets the maximum usable q (Table 4).

Results and Discussion

What drives the clustering

Table 1 and Figure 1a show that all three runs encode the same physical factors, with crystallinity and scattered intensity strongest. Causally (Table 2, Figure 1b), normalising the scattered intensity always degrades the map (ARI 0.01–0.11, collapsing entirely for SI3-log), and azimuthal averaging (*radial_only*) always collapses it to ~ 3 –6 classes (ARI ≤ 0.11): the decision needs the *two-dimensional* pattern, not a 1-D radial profile. Masking the low- q band is at least as damaging as masking high- q in every run, and dominates where the crop leaves little high- q content (SI4: 0.25 vs. 0.58).

Table 1: Representational probing (probe R^2 / η^2).

factor	SI4	SI3 lin	SI3 log
scattered intensity	0.70 / 0.23	0.59 / 0.39	0.84 / 0.61
crystallinity	0.80 / 0.57	0.57 / 0.46	0.83 / 0.60
spottiness	0.61 / 0.29	0.54 / 0.40	0.82 / 0.59

Table 2: Causal ablations (ARI vs. original map; K active).

ablation	SI4	SI3 lin	SI3 log
baseline	1.00 (9)	1.00 (12)	1.00 (14)
scattered_norm	0.01 (8)	0.11 (12)	0.00 (1)
radial_only	0.11 (6)	0.08 (3)	0.07 (3)
qmask low	0.25	0.32	0.27
qmask high	0.58	0.28	0.39

The DINO partition is not reproduced by classical methods

Table 3 and Figures 1c and 2 are the central result. No single classical feature set—and not their combination—reaches even ARI 0.18 against the DINO map, in any run, despite modest mutual information (AMI ≈ 0.2 –0.3). Visually (Figure 2), the classical clusterings recover the gross regions but produce blocky boundaries and miss the fine, sharp-edged grain structure that DINO resolves. Crucially, this is *not* because DINO uses exotic information—the ablations show it depends on the same scattered-intensity and low- q signals a classical pipeline measures—but because it combines them in a non-linear, in-plane-rotation-invariant way that a linear decomposition plus k -means does not reproduce.

Effect of the image stretch

Comparing the two SI3 runs isolates the pre-processing effect. The logarithmic stretch does not change *what* the model keys on—both runs collapse under azimuthal averaging and resist classical reproduction—but it sharpens it: K rises from 12 to 14, the physical factors become far more linearly decodable (probe $R^2 \sim 0.56 \rightarrow 0.83$), and the dependence on

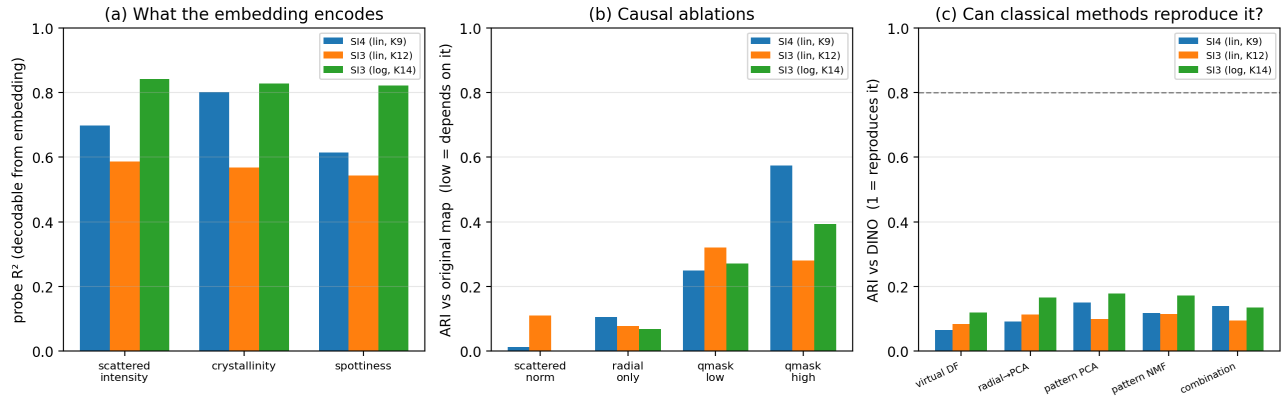


Figure 1: Cross-sample summary. (a) Physical factors are encoded in every run, most strongly under the log stretch. (b) Causal ablations: low ARI = the model depends on that signal; scattered-intensity normalisation and azimuthal averaging are the most damaging everywhere. (c) No classical feature set reproduces the DINO map (all $\text{ARI} \leq 0.18$; dashed line = 0.8 “reproduced” threshold).

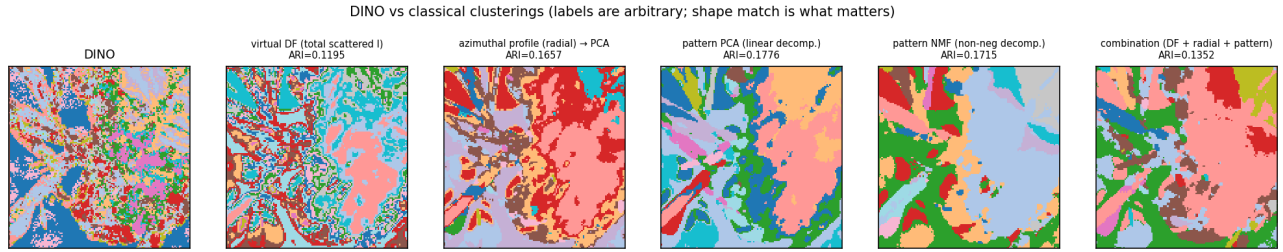


Figure 2: DINO vs. classical clusterings (SI3, log stretch; cluster labels are arbitrary, spatial shape match is what matters). Classical methods recover the coarse regions but miss DINO’s fine grain structure (ARI 0.12–0.18).

Table 3: Can a classical method reproduce the DINO map? (ARI vs. DINO; 1 = identical.)

classical feature	SI4	SI3 lin	SI3 log
virtual dark-field	0.07	0.08	0.12
radial → PCA	0.09	0.11	0.17
pattern PCA	0.15	0.10	0.18
pattern NMF	0.12	0.11	0.17
combination	0.14	0.09	0.14
best	0.15	0.11	0.18

the overall scattered intensity becomes extreme (scattered-norm collapses the log-stretch map to a single cluster, versus a reordered 12-class map under the linear stretch). Log compression linearises the intensity→feature relationship and lets the model resolve finer, more numerous classes around the same scattered-intensity / low- q axis.

Spatial-frequency window with the crop

Table 4: Effective q window. Low- q mask boundary is crop-independent ($q \simeq 2.1 \text{ nm}^{-1}$, $d \simeq 4.7 \text{ Å}$); the crop caps the maximum q .

sample	crop (px)	q_{max} (nm^{-1})	d_{min} (Å)
SI4	120	2.96	3.4
SI3	140	3.45	2.9

The tighter SI4 crop leaves little signal above the 2.1 nm^{-1} low- q boundary, which is why high- q masking barely perturbs SI4 (Table 2) while it matters more for the wider-crop SI3.

Conclusions

Across two IMC films and two intensity normalisations, the self-supervised 4D-STEM clustering rests on the overall scattered intensity plus the two-dimensional low- q ($d \gtrsim 4.7$ Å) diffraction structure, is invariant to in-plane rotation, and—most importantly—is **not a re-labelling of any classical virtual-DF, azimuthal, PCA, or NMF map** (best classical ARI ≤ 0.18). The DINO partition is therefore a distinctive, non-linear combination of otherwise-classical descriptors rather than a reproduction of an existing method. The logarithmic intensity stretch is the more discriminative preprocessing: it preserves this physical basis while resolving more, cleaner classes. Orientation/phase factors and a DINO-vs-zone-axis cross-check will be folded in once ACOM indexing (a CIF) is available; the present, CIF-free conclusions stand without it.

Acknowledgement Interpretability analyses were run on frozen, previously trained checkpoints.

References

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